

Homework (Jalapeno)

- Q1.** In ancient Egypt, a pharaoh wanted to build a pyramid with 2^3 times more bricks than the previous one. How many bricks will the new pyramid have if the previous one had 12 bricks?
(A) 48
(B) 24
(C) 96
(D) 64
(E) 32
- Q2.** Simplify the expression $x^2 \cdot x^4$
(A) x^6
(B) x^8
(C) x^{10}
(D) x^{12}
(E) x^{14}
- Q3.** Ancient Greek mathematicians used exponents to describe the growth of populations. If a population grows by a factor of 3^2 each year, how many times larger will it be after 2 years?
(A) 9
(B) 18
(C) 27
(D) 36
(E) 81
- Q4.** The Rosetta Stone has 2^5 characters inscribed on it. How many characters are there in total?
(A) 20
(B) 25
(C) 30
(D) 32
- (E) 40
- Q5.** The Great Wall of China has 3^2 watchtowers, each with 2^3 guards. How many guards are there in total?
(A) 72
(B) 64
(C) 48
(D) 36
(E) 24
- Q6.** The ancient Babylonians used exponents to calculate the area of triangles. Simplify the expression $(x^2)^3$
(A) x^5
(B) x^6
(C) x^8
(D) x^{10}
(E) x^{12}
- Q7.** In ancient Rome, a general had 2^4 soldiers, each with 3^2 shields. How many shields were there in total?
(A) 144
(B) 128
(C) 96
(D) 64
(E) 48
- Q8.** The ancient Mayans used exponents to describe the growth of their cities. If a city grows by a factor of 2^3 each year, how many times larger will it be after 3 years?
(A) 64
(B) 128
(C) 256
(D) 512
(E) 1024

Open Ended Questions (FRQ)

- Q9.** A ancient civilization built a series of temples, with each temple having 2^n more pillars than the previous one. If the first temple had 4 pillars, how many pillars will the 5th temple have?
- Q10.** Simplify the expression $\frac{x^4}{x^2}$ and explain the reasoning behind your answer

Chef's Tips

- Tip 1: Encourage students to use historical examples to illustrate their understanding of exponents
- Tip 2: Remind students to simplify expressions by applying exponent rules
- Tip 3: Emphasize the importance of clear and concise explanations in free-response questions